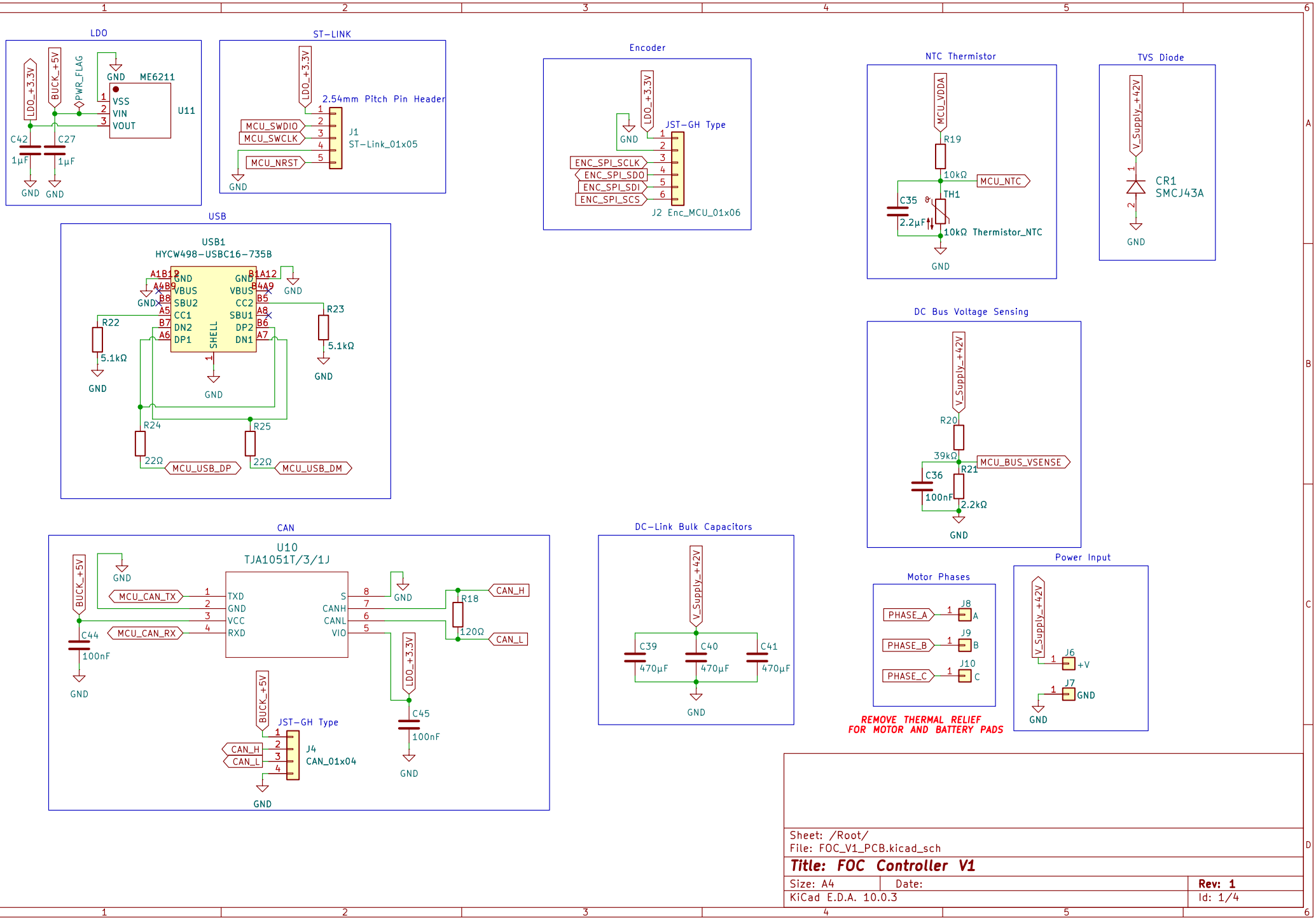
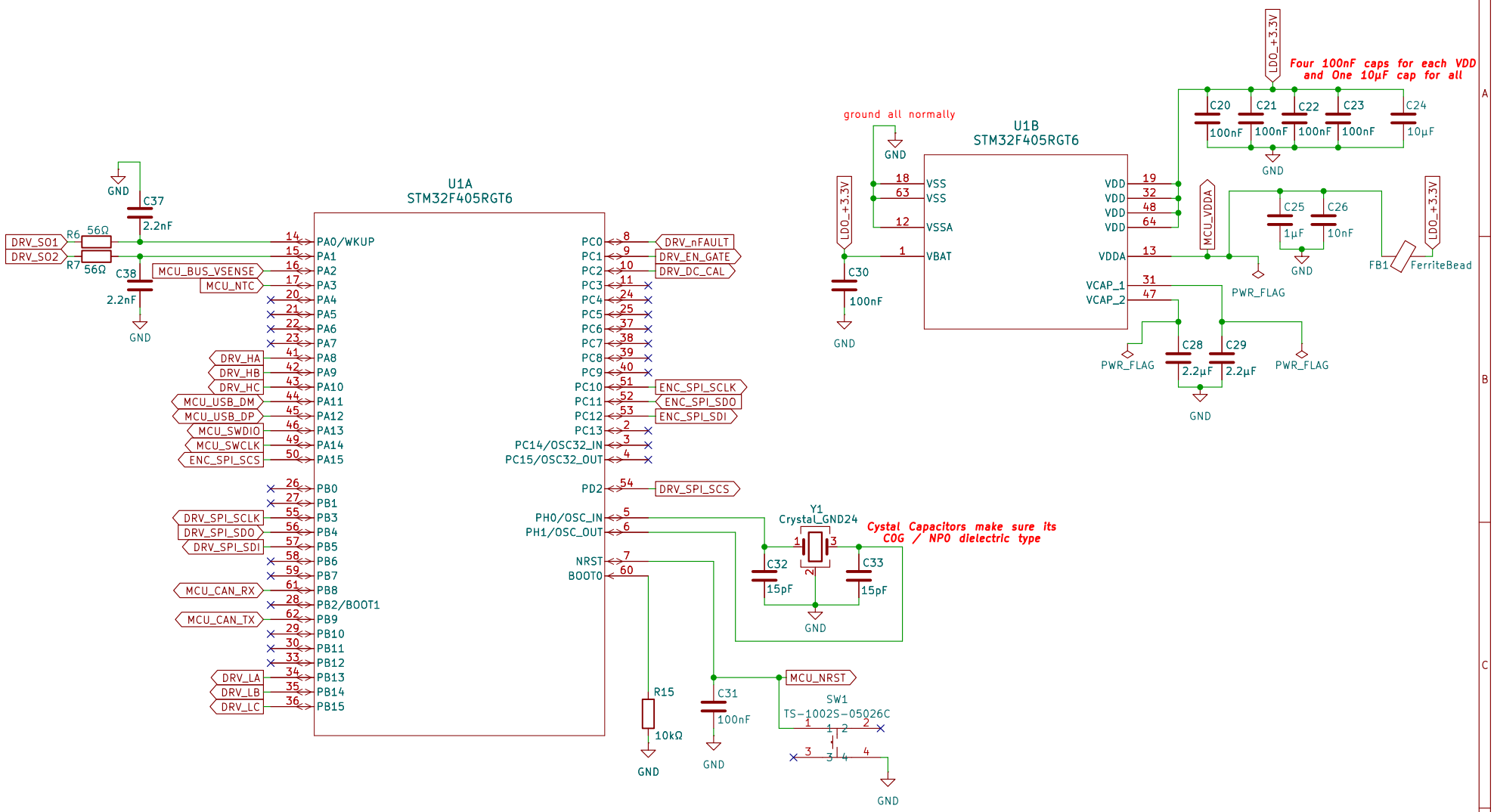
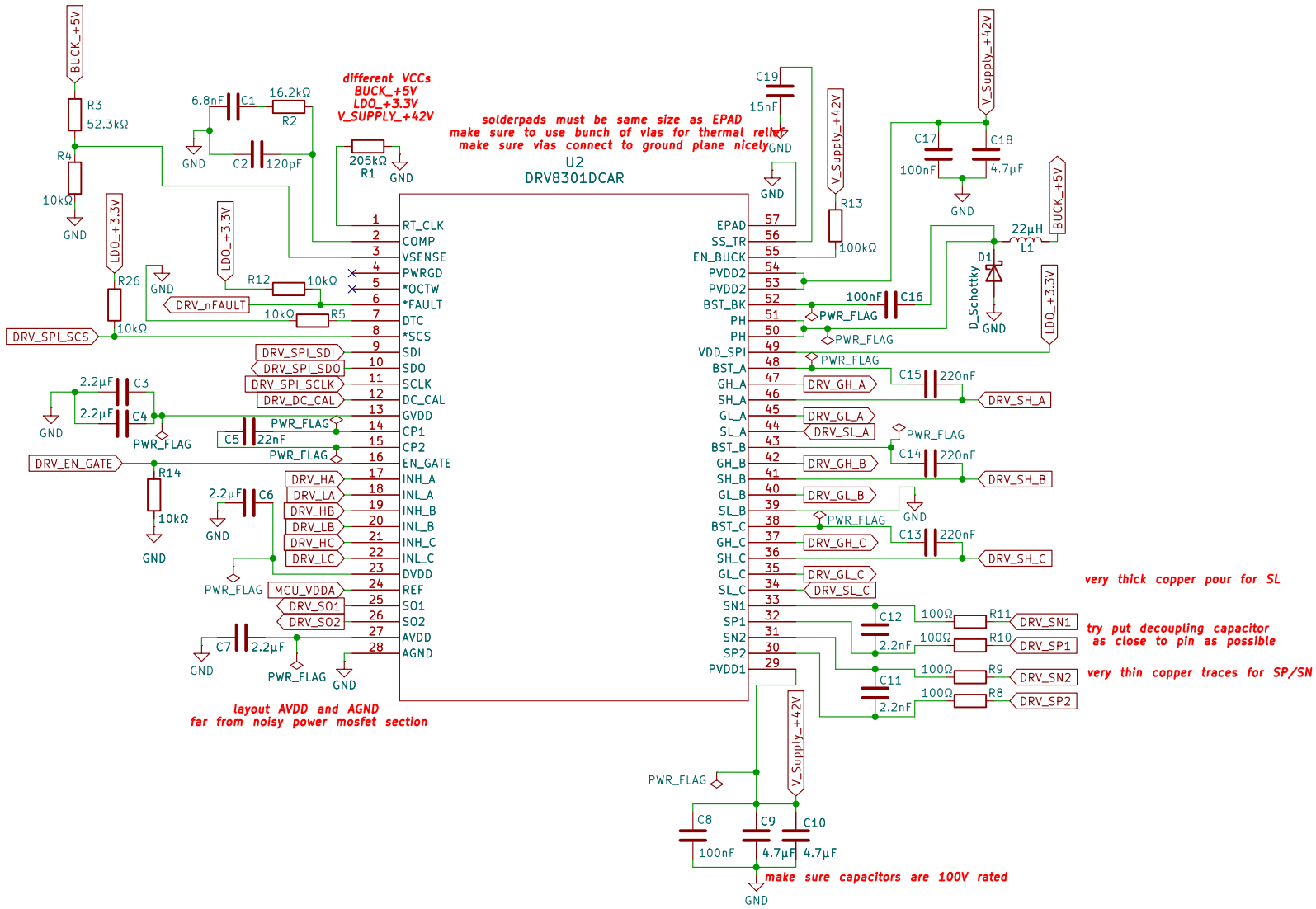
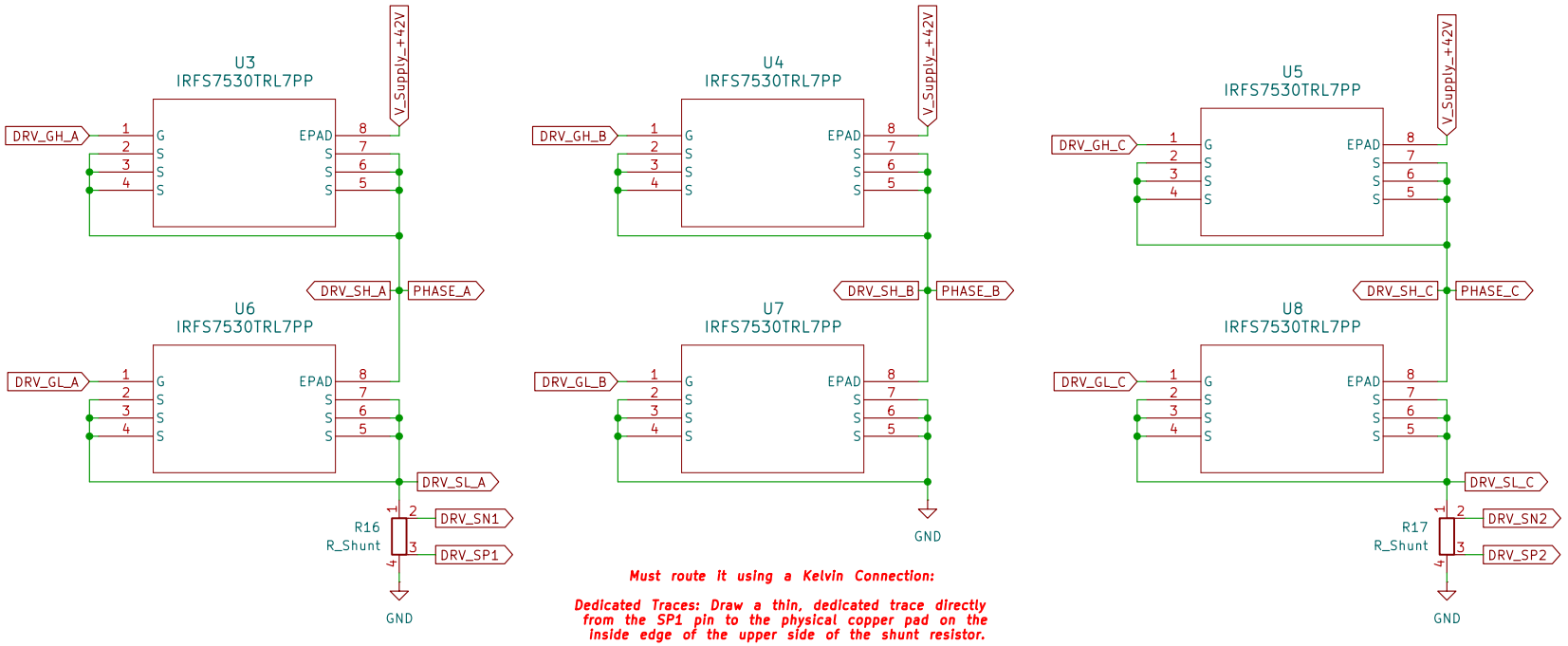

LDO
LDO_+3.3V
BUCK_+5V
PWR_FLAG
GND
ME6211
U11
VSS
VIN
VOUT
C42
1µF
C27
1µF
GND GND
ST-LINK
LDO_+3.3V
2.54mm Pitch Pin Header
MCU_SWDIO
MCU_SWCLK
MCU_NRST
J1
ST-Link_01x05
GND
Encoder
LDO_+3.3V
GND
JST-GH Type
ENC_SPI_SCLK
ENC_SPI_SDO
ENC_SPI_SDI
ENC_SPI_SCS
J2 Enc_MCU_01x06
NTC Thermistor
MCU_VDDA
R19
10kΩ
MCU_NTC
TH1
C35
2.2µF
10kΩ Thermistor_NTC
GND
TVS Diode
V_Supply_+42V
CR1
SMCJ43A
GND
USB
USB1
HYCW498-USBC16-735B
A1B12
A4B9
B1A12
B4A9
GND
VBUS
SBU2
CC1
DN2
DP1
SHELL
CC2
SBU1
DP2
DN1
B8
A5
B7
A6
B5
A8
B6
A7
R22
5.1kΩ
R23
5.1kΩ
R24
22Ω
R25
22Ω
MCU_USB_DP
MCU_USB_DM
DC Bus Voltage Sensing
V_Supply_+42V
R20
39kΩ
MCU_BUS_VSENSE
R21
C36
100nF
2.2kΩ
GND
CAN
U10
TJA1051T/3/1J
BUCK_+5V
MCU_CAN_TX
MCU_CAN_RX
TXD
GND
VCC
RXD
S
CANH
CANL
VIO
R18
120Ω
CAN_H
CAN_L
C44
100nF
LDO_+3.3V
C45
100nF
JST-GH Type
J4
CAN_01x04
DC-Link Bulk Capacitors
V_Supply_+42V
C39
470µF
C40
470µF
C41
470µF
GND
Motor Phases
PHASE_A
PHASE_B
PHASE_C
J8
J9
J10
A
B
C
REMOVE THERMAL RELIEF
FOR MOTOR AND BATTERY PADS
Power Input
V_Supply_+42V
J6
+V
J7
GND
GND
Sheet: /Root/
File: FOC_V1_PCB.kicad_sch
Title: FOC Controller V1
Size: A4
Date:
Rev: 1
KiCad E.D.A. 10.0.3
Id: 1/4



Sheet: /Microcontroller/		
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 2/4



Sheet: /Gate Driver/		
File: gate_driver.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 3/4



Must route it using a Kelvin Connection:

Dedicated Traces: Draw a thin, dedicated trace directly from the SP1 pin to the physical copper pad on the inside edge of the upper side of the shunt resistor.

Do the same for Ground: Draw another thin, dedicated trace directly from the SN1 pin to the physical copper pad on the inside edge of the bottom side of the shunt resistor. Do NOT connect SN1 to your main ground plane anywhere else it must only touch ground right at the resistor pad.

Route them together: Route the SP1 and SN1 traces right next to each other, parallel, all the way from the DRV8301 to the resistor. This is called a Differential Pair. Because they are right next to each other, any magnetic noise from the motor will hit both wires equally, and the DRV8301's amplifier will magically cancel the noise out!

Sheet: /Power Mosfet/
File: power_mosfet.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.3 | Id: 4/4